

Introduction

What we are trying to do is learn if models can perform better on OOD tasks through better prompting or if there is some way to elicit performance on OOD through priming of externally related concepts.

There are many ways we might help a model do this, but we are looking at hard prompting and looking if there are any prompt attributes that help us elicit better performance on OOD tasks?

We expect that although it will be unlikely to get completely accurate results from different prompts, but we are interested in seeing how close we might be able to get. We could think of this with the wine glass example as accuracy is reflected in the wine being as close to the top of the glass as possible.

There is an interesting dimension in performance on OOD in how they compare to human reasoning. Out of distribution tasks highlight a difference in how models can conceptualize things compared to humans. It raises further questions about how strong performance reflects understanding or not.

Experimental setup

Step 1: Determining an OOD Object

- Membership Inference Method
 - Based on method from *GenAI Confessions: Black-box Membership Inference for Generative Image Models*
 - Allows us to determine if a model has seen an object during training

Step 2: Prompt Stratification and Image Generation

- Determine different variables that could differ between prompts
- Use OpenAI's gpt-5.1 model to come up with prompts stratified across variables to have a large selection of variables
- Feed prompt to image generation model to get performance on task

Step 3: Evaluation with VQAScore

- The VQAScore that we are using is based on *Evaluating and Improving Compositional Text-to-Visual Generation* where it calculates the text to image alignment score.
- We use the VQAScore with gpt-4.1 in order to calculate the performance of prompts on the OOD task
- If certain tasks can get higher scores, it indicates that they elicited better performance on the task

Summary + Conclusion

Main Takeaways

- Performance on OOD tasks vary greatly, the same model has shown that it can perform, but not consistently
- Membership inference becomes a practical tool for OOD filtering
- VQA-based evaluation reveals subtle failure modes
- Certain attributes might improve performance on OOD, further testing is needed to clarify this trend

Conclusion

- Prompt stratification appears to influence generative model's ability to perform OOD tasks.
- While perfect accuracy remains difficult, structured and simplified prompting may provide measurable improvements and suggest ways in how models can be guided toward better generalization.

Limitations

- The sample size of prompts and models are finite
- VQA-Score is an imperfect proxy for “understanding”
- Membership inference threshold is unstable

Background

You may or may not have seen instances of image generation models not be able to perform on seemly simple tasks, like depicting a wine glass full to the brim with wine.

These kinds of tasks are called out of distribution (OOD), and models struggle with them because while a full glass of wine is a concept we can imagine, in reality there are relatively few instances, let alone pictures of these objects for models to learn.



“a glass of red wine that is filled completely to the brim”



“an empty stick of deodorant”

Results

Predictor	Estimate	p-value	Interpretation
word_count	0.08648	0.03690	Using a higher word count helps to increase VQA Score.
descriptor_words	-0.04787	0.02067	Using more descriptive words slightly decreases VQA Score
Spatial_constraints (difference from None) - Bottom - Side - top	- 0.32640 - 0.37015 - 0.14043	- 2.91e-06 - 2.16e-07 - 0.037037	Using spatial constraints compared to no spatial constraints have different effects in both directions.
num_visual_attributes	-0.07842	0.00172	Using more visual attributes also slightly decreases VQA Score
ambiguity	0.02191	0.816709	Ambiguity did not have any statistical significance.

Future directions

With the VQAScore, membership inference, prompting and image generation pipelines set up, we want to extrapolate this method to many more tasks and a bigger number of prompts.

Using more diverse prompts and stratifying along different variables.

There might be a difference between tasks, for example OpenAI took notice of the wine glass issue, and made tweaks to their gpt-4.1 models to perform better on this. While it still doesn’t perform great and was worth testing, results might differ between tasks as well.

References

VQA Score: Li, B., Lin, Z., Pathak, D., Li, J., Fei, Y., Wu, K., Xia, X., Zhang, P., Neubig, G., & Ramanan, D. (2024). *Evaluating and improving compositional text-to-visual generation*. 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 5290–5301. <https://doi.org/10.1109/CVPRW63382.2024.00538>

Membership Inference: Bohacek, M., & Farid, H. (2025). *GenAI Confessions: Black-box Membership Inference for Generative Image Models*. <https://arxiv.org/abs/2501.06399>